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Avadi, Chennai

**SCHOOL OF COMPUTING**
**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**
**Academic year 2025-2026**
**(Winter Semester)**

**Course code: 10214AD701** **Course Name: Major Project**

| TTS NO | Supervisor Name | VTU NO. | Student Name1 | VTU NO. | Student Name 2 |
|---|---|---|---|---|---|
| 3781 | Dr. N. Krishnammal | 23403 | J. Navadeep | 23598 | B. Sai Sumanth |

**Title of the Project: A Hybrid Approach for Predictive Waste Classification and Resource Optimization using Deep Transfer Learning and Reinforcement Learning**

**Problem Statement**

Rapid urbanization and industrial growth have led to massive increases in solid waste generation. Traditional waste management systems rely on manual sorting and static routing, leading to low recycling efficiency, high landfill overflow, loss of reusable resources and increased carbon emissions.

**Mapping of the Project to Complex Engineering Problem (CEP)**

| Level | Attribute | Characteristics | Targeted WKs (wherever applicable) | Justification (as applicable) |
|---|---|---|---|---|
| WP1 | Depth of Knowledge | Cannot be resolved without in-depth engineering knowledge at the level of K3, K4, K5, K6 or K8 | WK3, WK4, WK5, WK6, WK8 | The proposed system requires strong knowledge of deep learning (CNNs, transfer learning using ResNet50), reinforcement learning (MDP, PPO), and data preprocessing techniques. Integration of perception and decision-making modules demands understanding of AI system |

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| | | | | design, optimization strategies, and real-world waste management constraints. |
| WP2 | Range of conflicting requirements | Involve wide-ranging and/or conflicting technical and non-technical issues | WK4, WK5, WK6, WK8 | The system addresses trade-offs between classification accuracy and computational efficiency, as well as optimization between recycling benefits and operational costs. It balances model performance, real-time decision-making, and scalability while ensuring environmental sustainability and practical deployment feasibility. |
| WP3 | Depth of analysis required | Have no obvious solution and require originality in analysis | WK3, WK4, WK5, WK8 | The work involves detailed analysis of model selection, feature extraction, and reinforcement learning policy optimization. It requires evaluation of classification performance, reward design, and system integration, ensuring effective coordination between learning-based perception and decision-making components. |
| WP4 | Familiarity of issues | Involve infrequently encountered or novel problems | WK4, WK8 | The integration of transfer learning with reinforcement learning for waste management presents a novel and less-explored problem. It involves handling dynamic environments, uncertain inputs, and adaptive decision-making. |
| WP5 | Extent of applicable codes | Address problems not encompassed by standards and codes of practice | WK6, WK8 | Existing waste management systems and AI models do not fully address integrated classification and optimization. This project applies research-driven approaches combining deep learning and reinforcement learning, requiring custom |

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| | | | | model design, simulation environments, and performance evaluation. |
| WP6 | Stakeholder involvement and conflicting requirements | Involve collaboration across disciplines and diverse stakeholders | WK5, WK6, WK7 | The system involves multiple stakeholders such as municipal authorities, waste management agencies, and end users. It addresses requirements related to efficient waste segregation, optimized collection, and sustainable resource utilization through AI-driven decision support. |
| WP7 | Interdependence | Address high-level problems requiring a systems approach | WK3, WK5, WK6 | The proposed framework integrates classification and optimization modules, where performance of one directly impacts the other. It considers system reliability, scalability, and environmental sustainability, ensuring robust operation under dynamic real-world conditions. |

Mapping of the Project to Complex Engineering Activities (CEA)

| EA No. | Attribute | Characteristics | Justification based on the Project |
|---|---|---|---|
| EA1 | Range of Resources | Diverse resources and technologies | Applied concepts of deep learning, transfer learning, and reinforcement learning to design an intelligent waste management system integrating classification and decision-making. |
| EA2 | Level of Interactions | Optimization of complex interactions | Analyzed inefficiencies in traditional waste management systems and formulated the problem as a classification and sequential decision-making task. |
| EA3 | Innovation | Creative and research-based solutions | Designed a hybrid framework combining ResNet50-based classification and reinforcement learning-based optimization for efficient waste handling. |
| EA4 | Consequences to Society and Environment | Significant societal impact | Conducted experiments to evaluate model performance using classification metrics and reinforcement learning reward analysis in a simulated environment. |
| EA5 | Familiarity | Extends beyond prior experience | Utilized modern tools such as TensorFlow, Stable-Baselines3, Gymnasium, and Streamlit for model development and system integration. |

stainable Development Goals (SDG) Mapping

| SDG No. | SDG Title | Relevance to the Project |
|---|---|---|
| SDG 16 | Responsible Consumption and Production | Promotes efficient resource utilization through AI-driven waste classification and recycling optimization. |
| SDG 11 | Sustainable Cities and Communities | Enables smart waste segregation and optimized collection, improving urban sustainability and reducing environmental impact. |
| SDG 3 | Climate Action and Well-being | Reduces landfill usage and carbon emissions through optimized waste handling strategies. |

**Overall Justification:** The project addresses a real-world complex problem by integrating deep learning and reinforcement learning for intelligent waste management. It involves multidisciplinary knowledge, system-level design, and optimization under dynamic conditions, aligning with CEP and CEA requirements.

**Depth of Knowledge – Required Knowledge Levels (K3–K8)**

The problem addressed in this project cannot be resolved without in-depth engineering knowledge spanning the following **Knowledge Profile levels:**

**K3 – Engineering Fundamentals**

- Understanding of machine learning basics, image processing, and data preprocessing techniques for waste classification.

**K4 – Specialist Knowledge**

- Advanced knowledge of deep learning (CNNs, ResNet50) and reinforcement learning (MDP, PPO/DQN) for intelligent decision-making.

**K5 – Engineering Design**

- Design of an integrated system combining classification and optimization modules for efficient waste management.

**K6 – Engineering Practice**

- Practical implementation using tools such as TensorFlow, Stable-Baselines3, and Streamlit for model development and deployment.

**K7 – Engineering Comprehension**

- Ability to analyze system performance using evaluation metrics and interpret optimization outcomes for decision-making.

**K8 – Research Literature and Emerging Technologies**

- Utilizes recent research in transfer learning and reinforcement learning for intelligent waste management, and adapts emerging AI techniques to develop an integrated and efficient optimization framework.

N.V 30/3/2026

Signature

Dr. N. KRISHNAMMAL

(Supervisor Name)